Development and Evaluation of a Virtual Reality Snow Squall Driving Simulation

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Statements and Declarations
Conflicts of interest: The authors have no competing interests to declare that are relevant to the content of this article.

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Author Contributions
All authors participated in surveying activities and writing and editing the manuscript. Frank Martin designed and developed the simulation and assisted with surveying. Jase Bernhardt developed the surveys and led data collection. John Banghoff provided technical expertise on designing a scientifically realistic simulation and assisted with surveying. Sasha Pesci translated the simulation and survey results into Spanish and assisted with surveying.

Availability
The final version of the simulation can be found on the Meta Quest Store at the following link:
https://www.meta.com/experiences/snow-squall-driving-simulation/9642051659157182/
In addition, a browser-based 2D version of the simulation can be tried at:
https://weathersimulations.hofstra.edu/SnowSquall/index.html

Data Availability
All data analyzed during this study are included in the supplementary files.

Abstract

Snow squalls remain a dangerous meteorological hazard, particularly to highway drivers. While snow squalls do not cause significant amounts of total snowfall, their rapid onset and intense nature create an environment conducive to roadway fatalities. In particular, the near-zero visibility and slick roadways caused by snow squalls can catch drivers by surprise, leading to multi-vehicle pileups. Though the National Weather Service began issuing Snow Squall Warnings to mobile phones in 2018, many individuals remain unaware of the risks posed by this hazard. Additionally, extra cultural and language barriers exist when using traditional text-based warnings and guidance to communicate risks to Spanish-speaking groups with Limited English Proficiency. To enhance more traditional methods of risk communication, especially among non-native English-speaking groups, we created a virtual reality (VR) snow squall driving simulation. The simulation provides an open-ended environment in which users drive through a simulated snow squall. We conducted a user study in which 216 individuals tested the simulation and completed a survey. 146 of the participants used the simulation and answered the survey questions in English, while the other 70 participants used the simulation and completed the survey in Spanish. Using the results, we compared the effectiveness of the VR simulation across the two survey cohorts. We found the VR simulation to be an effective communication tool across both groups, with participants finding the experience enjoyable and reporting intent for behavior change if faced with a real snow squall.

1 Introduction

Recent years have seen the development and release of a multitude of immersive technologies in the sphere of augmented reality (AR), mixed reality (MR), and virtual reality (VR). The proliferation of VR headsets as household devices has continued since the release of the Oculus Rift and HTC Vive in 2016, with more recent products focused on providing an all-in-one untethered experience in which all aspects required for a convincingly immersive VR experience are handled by the chipsets and on board sensors within the headset itself [1]. More recently, VR technology has shown promise as a tool for simulation and visualization [2–4]. VR has also seen use in training and education for a variety of fields, such as for driving, first responders, medical procedures, and even military personnel [5]. This research aims to explore the usage of VR as a method for education and training, specifically for driving during a snow squall. Snow squalls are one of the most dangerous forms of winter weather in the northeastern region of the United States [6]. Therefore, there is significant value in the development of a simulation that allows for safe and effective education regarding snow squalls as a meteorological hazard.

1.1 Usage of Immersive Technology in Education and Training

Immersive technology has been increasingly studied as an educational medium and examined for its effects on information retention [7]. Pollard et al. [8] directly compared the impact of highly immersive learning environments in a head-mounted display (HMD) against more traditional forms of educational media and found that immersive environments can have a positive impact on knowledge retention. Learning in virtual environments generates interest and motivation among students [9], further pointing towards its effectiveness in the educational sector. It also has the added benefit of allowing learners to experience new and inaccessible environments thanks to the unique sense of presence granted by VR technology [9].

Particularly promising is the use of VR in simulated motor skills training with its unique ability to offer an affordable, consistent, repeatable, and safe environment for particular task completion. Many studies have used VR as a medium for task training with the goal of having the learned skills transfer to the physical world. Lie et al. [10] created an open-ended training simulation for drilling practice using a 3-axis milling machine. Similarly, Harvey et al. [11] developed and evaluated the potential of VR simulations for motor skill acquisition by developing a clay pigeon shooting simulation and directly compared the performance of beginners against experts at the activity. Many training simulation studies have been focused in the medical field, as these scenarios are often difficult to simulate in the real world without a physical patient being present, something which is not always practical. Examples include simulations of endotracheal intubation, patient examination and pre-operative planning, and cardio-pulmonary resuscitation [12–15]. Research in this area has demonstrated improved skill acquisition in medical students [16].

Prior work has examined the potential for VR to be used in disaster preparedness. These approaches often combine both educational and training aspects to create VR simulations that both relay knowledge and convey practical risk prevention behaviors. To test the potential of VR in disaster preparedness education, Frank et al. [17] developed a coastal storm surge simulation that allowed users to safely experience the threats of storm surges. The storm surge simulation was demonstrated to civic leaders to gather community-specific information on the usage of the application as an informational outreach tool [17]. A similar study was conducted using a simulation created to emulate an emergency flood evacuation, which took users through scenarios designed to teach safe evacuation behaviors [18]. Researchers evaluated the effectiveness of the flood simulation using questionnaires given to participants before and after using the application and found that knowledge of what to do during flood disasters increased significantly [18]. These prior studies demonstrate VR's effectiveness as a tool for education and preparedness training for high-risk events.

1.2 Overview of Snow Squalls

Snow squalls are defined by the US National Weather Service (NWS) as "brief bursts of heavy snow accompanied by gusty surface winds and characterized by a rapid onset of near-zero visibility" [19]. Though not resulting in high total amounts of snowfall, these events result in a significant loss of life [6]. Prior studies have linked general winter weather conditions to an increase in motor-vehicle-related fatalities. Roadway fatalities during winter weather conditions are highly correlated with changes in visibility, with 80% of fatalities being linked to decreases in visibility [20]. Highways are particularly vulnerable, with two-thirds of winter-weather-related fatalities occurring on these roadways [20]. Snow squalls remain a leading cause of weather-related fatalities, with squalls being the second leading cause of fatalities in Pennsylvania in recent years [19].

The NWS has made substantial efforts to warn drivers of hazardous winter weather by issuing event-driven Watches, Warnings, Advisories, or Special Weather Statements [6]. Despite the heavy snowfall caused by squalls and the associated impacts, squalls do not produce a large enough volume of snow accumulation to warrant issuance of Winter Weather Advisories. Additionally, the rapid onset of snow squalls makes driver adaptation difficult [21]. This has resulted in a disconnect between the dangers caused by snow squalls and the tools available to meteorologists to warn drivers. More recently, efforts have been made to create new warning systems to alert the general public about snow squall events. The NWS began delivering Snow Squall Warnings directly to mobile phones in 2018 using the Wireless Emergency Alert (WEA) system and the Pennsylvania Department of Transportation (PennDOT) and some states now deliver Snow Squall Warnings through changeable message signs (CMS) found along highways [22]. While the efforts to warn the public of inclement weather are important, the rapid onset of snow squalls means that drivers that are already travelling on the road may miss these warnings. Therefore, efforts to educate the general public about snow squalls and new methods to train appropriate accident mitigation measures could be an important method to reduce snow squall-related fatalities.

1.3 Existing Snow Squall Outreach and Education Efforts

Since the implementation of Snow Squall Warnings in 2018, the NWS has pursued a variety of outreach and education initiatives to improve public understanding and preparedness. One of the primary barriers to the success of snow squall outreach and education has always been a visual depiction of snow squalls and their resultant pileups. Just as public service announcements encouraging people to quit smoking illicit fear of continuing (or starting) the harmful behavior through their inclusion of individuals with raspy voices, a dramatic portrayal of rapidly changing road conditions or an aerial view of the aftermath of a chain-reaction pileup may also discourage people from thinking they can make it safely through a snow squall when a Warning is issued. Although numerous pictures of pileups exist, the cost for broad distribution is prohibitive and their contents are often too graphic. Meanwhile, the NWS does not possess the technology to develop digital depictions of snow squalls or their resultant pileups for a broad outreach campaign.

In February of 2021, a resident from Wisconsin offered up extensive 360° video footage from his Tesla as he drove through a snow squall. That footage was the final piece needed to launch a safety campaign. By the following November (2021), the NWS office in State College, PA designed and implemented an education campaign focused on snow squall science, warnings, and safety as part of an annual Snow Squall Awareness Week. The content, housed at www.weather.gov/ctp/snowsquall [23], was designed in partnership between the NWS and other Pennsylvania state partners including the following Pennsylvania agencies: Department of Transportation (PennDOT), Turnpike Commission (PTC), Emergency Management Agency (PEMA), and State Police (PSP). This joint effort resulted in a comprehensive frequently asked questions guide that covers everything from what a snow squall is to what to do if caught in a snow squall pileup [23]. Since then, Snow Squall Awareness Week has been held each year with content shared by more than 20 NWS offices across the northeast US to help raise awareness.

In 2022, NWS State College partnered with the University of Oklahoma Institute for Public Policy Research and Analysis (OU IPPRA) to add a few questions about snow squalls to their annual Extreme Weather and Society Survey [24, 25]. The intent of this collaboration was to gauge end user exposure to and familiarity with snow squalls, warnings, and recommended responses. Conducted each year, the results provide a cross-sectional snapshot of public awareness broken down by a variety of demographics. An analysis of responses to snow squall questions collected in 2022 and 2023, which each included nearly 1,500 respondents spanning the US, found several important results. First, nearly half of the respondents had never heard of a snow squall Warning. Given that survey participants span the U.S., the broad lack of familiarity with a relatively localized weather event is not surprising. Even still, anyone who drives through snow squall-prone areas needs to be aware of their existence no matter where they reside. Second, nearly 50% of respondents rated their understanding of snow squall warnings as “Poor” or “Fair” and open-ended responses indicated significant gaps in understanding of recommended mitigating practices when a snow squall Warning is issued.

The responses to the snow squall related questions on the Extreme Weather and Society Survey present a compelling case for continued and expanded efforts to educate the public about snow squalls and their dangers. NWS offices' primary avenue for outreach is via social media, and their reach is limited to X, Facebook, and Instagram. Furthermore, leveraging advanced technologies, such as VR, to improve visualization is a potential vector to foster greater outreach efforts. Thus, the intended benefits of a newly developed snow squall VR simulation were two-fold: (1) better visualization of the dangers of snow squalls and (2) broader exposure.

1.4 Information Access and Language Barriers

The Spanish-speaking population in the United States (U.S.) has seen continued growth. According to the 2023 U.S. Census, 13.9% of the total population speaks Spanish at home, making it the second most spoken language in the U.S. [26]. Spanish speakers in the U.S. with Limited English Proficiency[1] are often more vulnerable to weather-related disasters due to systemic, social, economic, and political inequalities [28]. Several disasters, such as the Saragosa Tornado in 1987 [29] and the Oklahoma tornadoes and flash floods in 2013 [30] have demonstrated the vulnerability of historically marginalized communities, such as those of Latin American and Hispanic ethnicity. To avert future disasters resulting in a loss of life among vulnerable populations, it is of particular importance to ensure accurate risk communication messaging reaches these groups [31]. Communication efforts often face additional challenges due to the diversity of backgrounds present among the Spanish-speaking population. Immigrant groups, particularly those with LEP, can be ill-equipped to handle geological and meteorological hazards that may not be present in their country of origin, and regional dialects can affect the interpretation of warning messages [32].

Thus, it is imperative to incorporate new and innovative strategies to risk communication that increase access across language barriers. VR simulations offer a promising new method to foster understanding of dangerous conditions and hazards. Yet the number of simulations that provide lifesaving hazardous weather information in languages other than English is limited. While the impact of traditional text-based communication methods may be limited by translation issues and a lack of prior experience with the hazard, VR simulated scenarios can offer a way to overcome language barriers.

1.5 Contributions

To reduce risks associated with snow squalls that occur as a result of incorrect driver decision-making and the inadequate education surrounding this hazard, we developed a VR snow squall driving simulation. Our application is an immersive visualization of a snow squall that allows users to drive through dangerous conditions without personal risk. This tool serves multiple purposes:

[1] Limited English Proficiency (LEP) is a term that the Census Bureau uses to refer to individuals who report speaking English less than "very well," based on their responses to a question in the American Community Survey [27].

- Provides a realistic depiction of a snow squall that can help drivers recognize the hazard and associated risks.
- Educates current and future drivers on what actions they can take to decrease the likelihood of accidents.
- Allows for analysis of the impact of VR simulations across English-speaking and Spanish-speaking populations.

In pursuit of these goals, we tested the simulation with the general public through a user study. We present the methodology used to develop the simulation and our analysis of the results in hopes to influence future work in the development of environmental hazard-related simulations.

2 Methods

2.1 Design Specifications

We began designing the simulation by first collecting insights from experts in the areas of meteorology and road safety in the northeastern region of the United States. Our expert contacts primarily were sourced from the NWS, PennDOT, and the New York State Weather Risk Communication Center. We held interviews with experts to collect their ideas and build a better understanding of the appearance of snow squalls and what aspects and events contribute to the associated unsafe driving conditions. Experts repeatedly mentioned the appearance of the impending snow squall as a wall of snow coming towards the driver and a swift reduction in visibility, paired with the sudden change from clear midday conditions to heavy snowfall. Groups from both the NWS and PennDOT also emphasized the danger of multi-car and truck pileups that occur due to the lack of visibility at high speeds of travel. While travelling at high speeds is dangerous, proceeding too slowly or stopping on the road also creates risks, as other drivers may not expect a stopped or nearly stopped vehicle on the road. NWS meteorologists specifically mentioned the flash-freezing effect on roadways, which causes drivers to underestimate the necessary braking distance. Although gusty winds often accompany snow squalls, meteorologists indicated that impacts from visibility reductions and flash freezing on roadways create a much higher threat to motorists. While the associated gusty winds may have a greater impact on larger passenger vehicles such as high-speed trains [33, 34], the impact on passenger vehicles during a snow squall is limited. As such, we focused on simulating visibility loss and roadway conditions associated with snow squalls and did not incorporate any wind effects in this simulation. Experts across all groups requested the inclusion of a Wireless Emergency Alert and a Changeable Message Sign, both of which are common methods that these agencies use to issue snow squall warnings [22].

Driving during snow squalls presents a complex set of dangers to the driver. Best practices to mitigate risk include not driving at all when a snow squall is imminent, and getting off at the nearest exit [23]. In reality, these risk avoidance practices are not always possible, and thus the simulation had to take into account a variety of other potential evasive actions, such as whether or not one should pull over and park on the side of the road and whether drivers can avoid a crash by driving slowly. This dynamism presents a challenge when accurately simulating such an experience. To combat these difficulties, we designed the simulation akin to an open-ended learning environment (OELE). Previous work presenting preparedness for environmental risks such as the storm surge danger simulation demonstrated in [17] adheres more closely to a traditional directed learning approach, in which the scenario designer is responsible for determining what information they deem important and the learner is taken through the information systematically, after which they are expected to recall and demonstrate the learned knowledge as it was presented [35]. In contrast, open-ended learning environments can be didactic in nature, allowing the learner to use their own cognition and unique intentions to solve problems within a larger context while engaging with the environment [35, 36].

2.2 Implementation

The simulation was developed for the Meta Quest 3 headset, which features a Snapdragon XR2 Gen 2 processor that delivers twice the graphical performance of the Snapdragon XR2 processor found in the Meta Quest 2 [37]. The extra processing power of the Quest 3 allowed us to render a high volume of snow particles without performance falling below 72 frames per second (FPS), which is the recommended minimum in order to prevent motion sickness [38]. We created the simulation using version 2022.3 of the Unity 3D game development engine.

In pursuit of an open-ended design, the user is placed behind the wheel of a vehicle on an open road with little guidance beyond a basic explanation of the VR controls. The terrain and environment were designed to resemble an interstate highway in Pennsylvania and New York. The user's car is controlled by the Touch controllers included with the Quest 3 headset. To steer, users can reach their virtual hands out towards the steering wheel and press the grip buttons down. Simulated hands realistically grip onto the steering wheel, which is rotated by controller movement in the real world. Beyond that, users can accelerate the vehicle by pressing the right Touch controller trigger button. Brake and reverse are mapped to the left Touch controller trigger button, akin to racing video games. Simulation controls were displayed to users in the application before beginning the driving experience through a simple diagram of the Touch controllers coupled with text boxes that outlined button functionality. The simulated car also features two working side-view mirrors and a rear-view mirror, implemented by placing invisible cameras on the car that display their field-of-view to render textures on the mirror surfaces. Scripts update each mirror's texture every third frame, alternating between them, to ensure stable performance.

We designed the simulation to allow users to traverse the route at their own pace, encountering scripted events until an end condition is met. The simulation does not inform the virtual reality driver of what actions they can take that will affect or end the simulation. An outline of the scripted encounters and exit conditions can be found in Fig. 1. The end conditions are as follows:

- Collided with another vehicle on the road
- Exited the highway
- Reached the pileup
- Pulled over on the side of the road

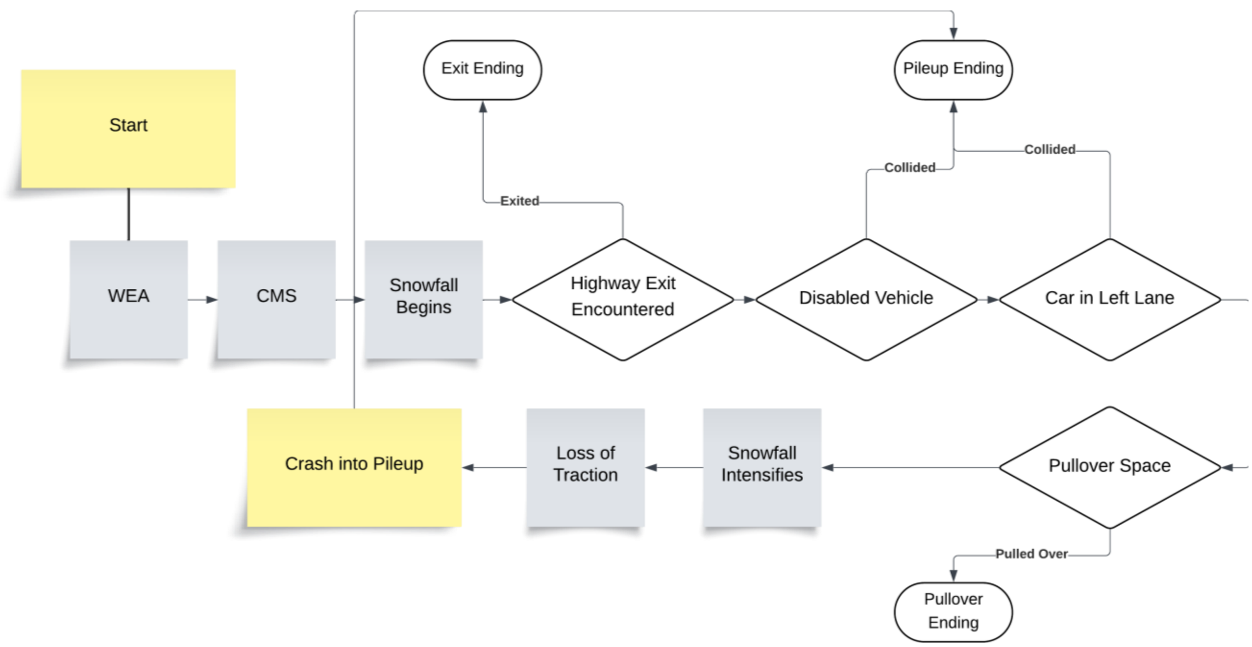

Fig. 1 General flow of the simulation.

2.3 Snow Appearance and Traction Physics

Snow squalls feature intense blowing snow that limits visibility [19]. Simulating reduced visibility during snow squalls through filling the scene with particles requires performance exceeding the capabilities of the Quest 3's hardware. Instead, the particle system follows the controlled vehicle and emits 400 particles per second over the windshield. Snow particles have a short lifetime of 0.3 seconds, are destroyed upon colliding with the vehicle, and do not collide with other objects in the scene. Poor visibility is primarily simulated using a textured box that fades from transparent to opaque and is scaled down in size over the course of 15 seconds until close to the vehicle. This effectively simulates the thick, foggy look of driving into a snow squall without negative performance impacts that could cause discomfort to users. Snow is applied to other surfaces in the scene via a decal-based approach. The snow decals are overlaid on top of other materials. The opacity and coverage of the decal is shifted over time, conveying the appearance of snow buildup on the roadway and surrounding terrain (Fig. 2).

Should the user continue driving along the road, they will encounter a large multi-vehicle pileup. Shortly before this, the user passes through an invisible trigger, causing the snow to rapidly build up until the road is no longer visible (Fig. 3). As the snow coverage on the roadway increases, a script modifies the vehicle's brake torque on both axels. This provides an approximation of the increased braking distance and loss of traction that can occur in a snow squall. If the user is traveling full speed, they will not be able to bring the vehicle to a complete stop even if they attempt to activate the brakes immediately when seeing the pileup.

Fig. 2 Snow decals on the roadway, snow particles, and visibility reduction.

2.4 Non-player Agents

Past studies have demonstrated that the inclusion of non-player characters (NPCs) in virtual environments is an effective way of engaging users and increasing the sense of presence and realism [4, 39]. While the simulation does not include any visible human characters, we did build several non-player agents into the simulation. These take the form of other vehicles and serve to increase realism and direct user behavior. The first and most simple agent is the disabled vehicle, which simply remains stationary in the right lane (Fig. 4a). Its only behavior is to flash its hazard lights, indicating the dangerous driving conditions to the user. The second is the speeding car, a red sports car that travels too quickly in the left lane (Fig. 4a). It follows a predetermined path using Unity's navigation system [40]. The user may struggle to see this car in the mirrors due to its fast speed and the poor visibility. If the user makes it to the pileup at the end of the simulation, they can identify this agent has slammed into the other cars and is unable to move. This demonstrates the dangers of not adapting one's speed and awareness to the weather conditions. Lastly, the chaser car (Fig. 4b) was implemented to emphasize the dangers of driving too slowly or coming to a stop on the road during a snow squall according to PennDOT. It appears if the user slows their vehicle substantially or the user's vehicle is stopped for any reason. The only exception to this is if the user drives their vehicle off of the road and stops in an area where there is a substantial and safe margin. Though this agent also uses Unity's navigation system, it does not follow a set path and instead follows behind the user's vehicle, slamming into it if it catches up. It also appears near the pileup area (Fig. 3) and will collide with the user's vehicle regardless of whether they manage to stop the simulated vehicle before the multi-car pileup. The simulation ends if the user's vehicle collides with any of these agents.

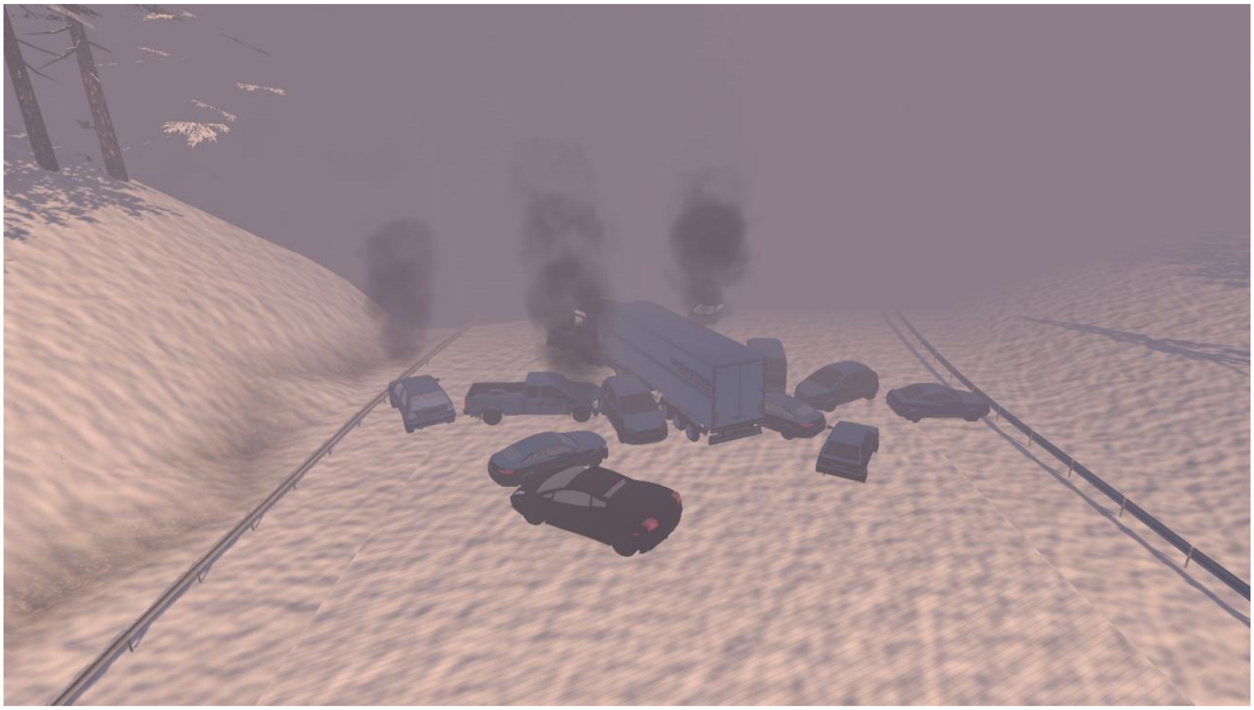

Fig. 3 Simulated multi-car pileup.

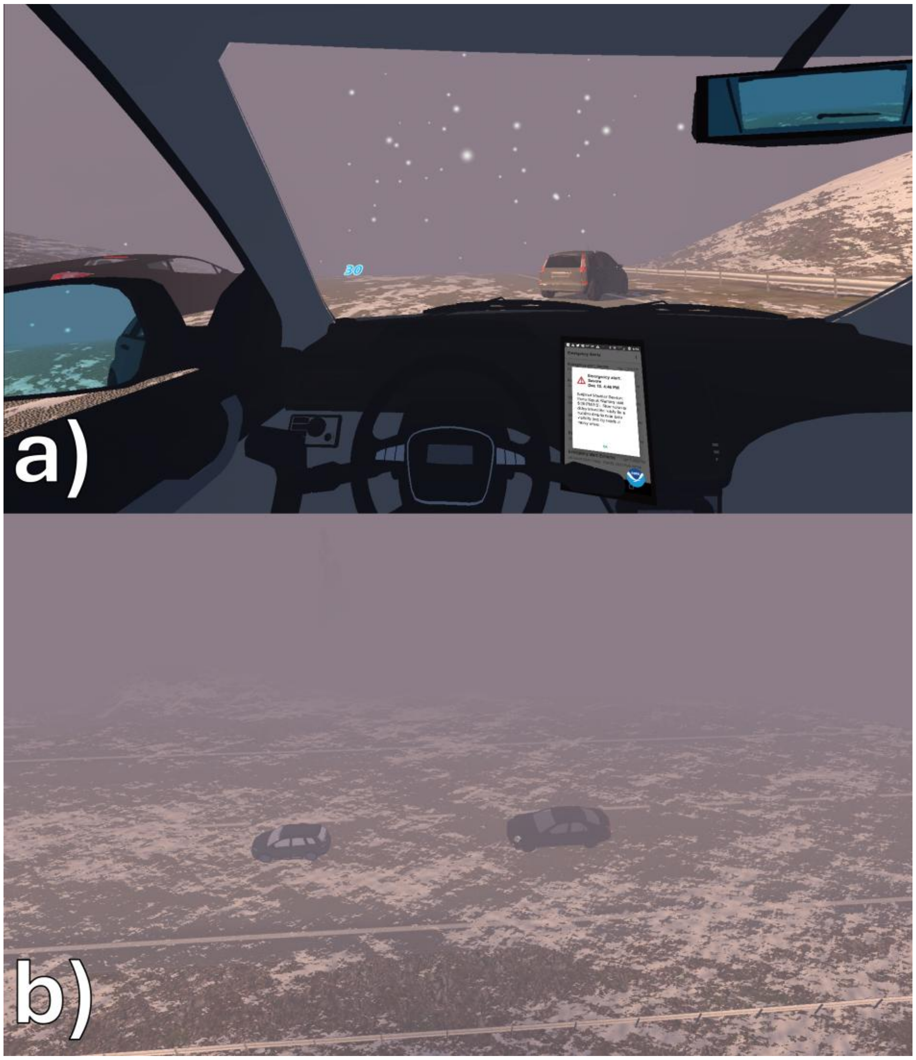

Fig. 4 Non-player agents. (a): The disabled vehicle in the right lane and speeding car in the left lane. (b): The chaser car approaching a stopped user vehicle.

2.5 Changeable Message Sign and Wireless Emergency Alert

The simulation also incorporates a Wireless Emergency Alert (WEA) (Fig. 5) and Changeable Message Sign (CMS) (Fig. 6) to enhance realism. These were created using the guidance of the NWS office in State College, Pennsylvania and PennDOT to ensure accuracy. Shortly after commencing the drive, a virtual phone, mounted near the car's dashboard, begins to beep and displays a snow squall warning. The sound and wording of the WEA match those used in real world alerts. Real WEAs can appear in both English and Spanish depending on the settings of a user's mobile phone. Our virtual phone uses the appropriate WEA matching the language set in the simulation prior to starting the scenario.

After the WEA, users will also encounter a CMS informing them of the risk of a potential snow squall. The text on the sign is taken directly from PennDOT's Changeable Message Sign

Operating Standards [41]. This text is not translated, as CMS text typically only appears in English alongside U.S. roads.

The WEA does not begin on a timer and instead starts playing shortly after the user begins driving down the road. This gives users a chance to familiarize themselves with the controls and environment before hearing the WEA. The WEA beeping sound effect plays for 11 seconds. The CMS can be seen ahead on the road and is passed shortly after the WEA activates. Both of these elements occur prior to the onset of snowfall, as outlined in Fig. 1.

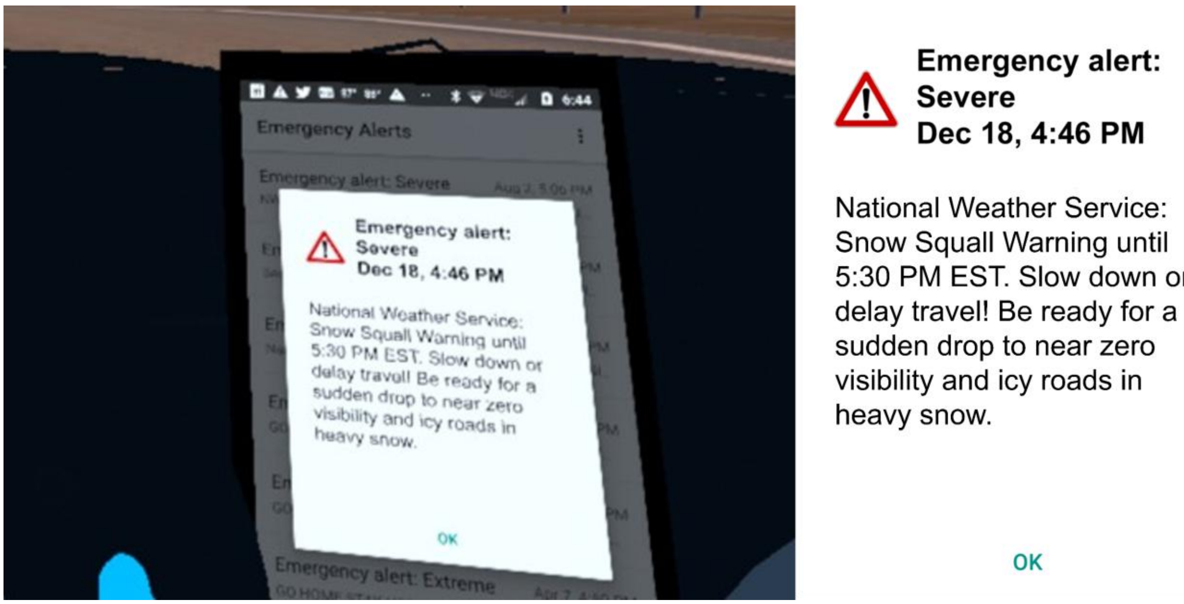

Fig. 5 WEA as it appears in the simulation.

Fig. 6 CMS as it appears in the simulation.

Upon the end of the simulation, users are shown a graphic corresponding to their exit condition. After the user meets an end condition, they then see a large poster in their field of view corresponding to the actions they took in the simulation, such as exiting the highway (Fig. 7a), pulling over and parking on the side of the road (Fig. 7b), or getting caught in a pileup (Fig. 7c). The pileup graphic is triggered on any exit condition in which the user collided with another vehicle. A final graphic, produced by the NWS and National Oceanic and Atmospheric Association (NOAA) is then displayed containing the message "There is no safe place on the highway during a snow squall," some information about snow squalls, and a link directing users to a weather.gov webpage from which the information is sourced (Fig. 7d) [23].

Fig. 7 Ending graphics. (a) The user exits the highway. (b) The user successfully parks on the side of the road. (c) The user participates in a collision. (d) Slide shown to all users after the end state graphics.

2.6 Survey Design and Data Collection

We tested the initial version of the simulation by conducting a user study on members of the public. The survey was created in Qualtrics and was available in both English and Spanish. Participants had the choice between taking the English or Spanish version of the survey. Those who chose Spanish are referred to as the Spanish-language cohort. This group included a mixture of bilingual and LEP Spanish-speaking participants, with 96% of that group speaking Spanish at home. Questions were designed to assess the impact of the VR simulation and its efficacy as a tool to incite behavioral change, as well its general usability, likability, and realism. Feedback regarding usability, likability, and realism was collected using a methodology similar to Maniker et al. [15]. Data was collected in the form of both qualitative written feedback and quantitative rankings.

Surveys were run across multiple locations in Pennsylvania and New York with varying amounts of snow squall warnings. Surveys were conducted between January and July 2025 across Long Island and Plattsburgh in New York and Pittsburgh and State College in Pennsylvania. A total of 216 individuals tried the VR simulation and took the survey. 146 individuals were surveyed in English and 70 were surveyed in Spanish. A bilingual researcher on the team helped to facilitate the survey and VR experience with the Spanish-language cohort. Survey results were translated into English for analysis.

A researcher was present with individuals while they engaged with the virtual reality simulation. The researcher first outlined the basic setup of the simulation, that they would be placed behind the wheel of a car, and explained the controls. The participant was then asked to put on the headset, where they had a second chance to review the control information via an in-application explanation screen. Researchers used the Meta Quest's streaming system to monitor each individual's progress through the simulation and offered additional assistance if the participant was struggling with the controls. This also allowed our team to closely monitor the feedback outside of what the individual noted in the survey.

3 Results

Participants were asked seven Likert-scale and two free-response questions to evaluate the VR simulation. Likert-scale responses were compared across the English- and Spanish-language cohorts, henceforth referred to as survey group 1 and survey group 2 respectively, using two-tailed t-tests ($\alpha = 0.05$). The first three questions assessed the effectiveness of the simulation as a potential tool for education and behavioral change. Participants provided answers on a scale from 1 to 5, with 1 indicating that the simulation was unhelpful/unimpactful and 5 indicating it was very helpful/impactful.

Two of the seven Likert-scale items showed statistically significant differences between groups. When asked "to what extent did this VR experience impact/change the way you might respond to a snow squall in the future?", survey group 2 reported significantly higher impact (survey group 2 mean = 4, survey group 1 mean = 3.66, $p = 0.042$). Similarly, survey group 2 also reported the simulation was significantly more helpful in showing what to do if caught in a snow squall compared to survey group 1 (survey group 2 mean = 4.13, survey group 1 mean = 3.69, $p = 0.009$).

The second four Likert-scale questions covered the self-reported ease of use, realism, and likability of the simulation. A score of 1 indicated a negative response (for example, the participant did not like the simulation), and a score of 5 indicated a positive response. Participants rated the simulation highly on all metrics, with the lowest mean score being ease of use at just below 4.00. Both groups responded very positively when asked whether they would recommend the simulation to their peers (survey group 1 mean = 4.32, survey group 2 mean = 4.31, $p = 0.056$). There were no statistically significant differences in the responses to any of the latter four questions between the two survey groups. Full results can be found in Table 1.

Table 1 Likert-scale question results, bold indicates significant differences of $p < 0.05$ as computed by a two-tailed t-test.

Question	**Survey group 1 (English) mean**	**Survey group 1 (English) SD**	**Survey group 2 (Spanish) mean**	**Survey group 2 (Spanish) SD**	***t***	***p***
To what extent did this VR experience impact/change the way you might respond to a snow squall in the future?	3.66	1.13	4.00	1.13	2.05	0.042
How helpful was the VR in showing you what a snow squall looks like?	3.86	1.17	4.16	0.99	1.82	0.070
How helpful was the VR in showing you what to do if you are caught in a snow squall?	3.69	1.16	4.13	1.08	2.65	0.009
The VR simulation was easy to use	3.95	1.17	3.72	1.26	1.28	0.200
The VR simulation was realistic	3.96	0.95	4.16	1.00	1.41	0.160

I liked the VR simulation	4.26	1.06	4.31	0.99	0.36	0.720
I would recommend using this VR simulation to my peers	4.32	0.99	4.31	0.91	0.06	1.000

Answers to the free-response questions that were submitted in Spanish were translated by a native Spanish speaker on our team. The first free-response question asked participants to explain their answers to any of the first three Likert-scale questions. Responses to this question were analyzed by our team and classified as positive, negative, or neutral in sentiment. A response could be categorized into multiple categories if it included multiple pieces of information with differing connotations. For example, one participant responded, "It mostly helped me figure out what to do in a snow squall because of the limited vision, but gauging the impact of snow and ice on traction was a bit difficult." This answer was considered both positive and negative due to the individual finding certain aspects of the simulation effective (the limited visibility) but others lacking (gauging the impact of poor traction was challenging). Of the 146 responses recorded from survey group 1, 131 were able to be scored in this manner. In survey group 2, of the 70 responses, 51 were able to be scored and 19 were left unscored. Responses were unscored due to being left blank or filled with text that lacked any discernible feedback such as "N/A." The amount of data labelled into positive and negative categories can be found in Tables 2 and 3. The data was compared across survey groups using two-proportion *z*-tests, which can be found in Table 4.

Participants had positive impressions regarding the simulation, with 70.23% of responders from survey group 1 and 86.27% of the responders from survey group 2 leaving positive comments. As seen in Table 2 and Table 3, positive-only comments were much more frequent than negative-only comments. The significance of this data was verified using McNemar's test (survey group 1 $\chi^2(1) = 29.58$, $p < 0.001$; survey group 2 $\chi^2(1) = 32.36$, $p < 0.001$).

This difference in positive sentiment was statistically significant between the survey groups, with survey group 2 overall having a higher proportion of positive feedback ($z = -2.24$, $p = 0.025$). Differences between the survey groups were more pronounced in negative feedback. 27.48% of the responses left by survey group 1 contained a negative aspect, much larger than the 9.08% of responses from survey group 2 who left similar feedback. This difference between the negative sentiment expressed by both groups was found to be significant ($z = 2.56$, $p = 0.010$). Very few scored responses were classified as neutral, with just 10.69% and 5.88% of responses from survey group 1 and survey group 2 falling into this category respectively. There was no significant difference between the proportions of neutral responses ($p = 0.317$).

Table 2 Distribution of positive-only and negative-only sentiment labels within the survey group 1 (English-language) responses. $\chi^2(1) = 29.58$, $p < 0.001$.

Survey group 1 (English) responses	Count
Positive only	81
Negative only	25
Both positive and negative	11
Neither	14

Table 3 Distribution of positive-only and negative-only sentiment labels within the survey group 2 (Spanish-language) responses. $\chi^2(1) = 32.36$, $p < 0.001$.

Survey group 2 (Spanish) responses	Count
Positive only	43
Negative only	4
Both positive and negative	1
Neither	3

Table 4 Free-response question results cross-group comparison, bold indicates significant differences of $p < 0.05$ as computed by a two-proportion z-test.

Category	Survey group 1 (English) count N=131	Survey group 2 (Spanish) count N=51	z	p
Positive	92 : 70.23%	44 : 86.27%	-2.24	0.025
Neutral	14 : 10.69%	3 : 5.88%	1.00	0.317
Negative	36 : 27.48%	5 : 9.80%	2.56	0.010

4 Discussion

In the present study, we developed a simulated experience of driving through a snow squall. Our goals were to create an application that helped drivers recognize snow squalls and to provide education in order to reduce the risk of automotive accidents. We also aimed to deliver this information effectively across both English and Spanish-speaking populations.

4.1 Analysis of the Survey Results

Results demonstrated that the simulation successfully created a recognizable depiction of a snow squall. Participants indicated that the simulation was helpful in demonstrating the visual appearance of a snow squall. In the qualitative feedback collected, multiple participants reported that the simulation was similar to snow squalls that they had previously experienced. Examples of this include comments such as “The VR simulation accurately represented my experiences in snow squalls before,” “It was a lot like what the snow squalls I’ve experienced have been like,” “The headset was very immersive. Realistic to my own experience,” and “VR was very similar compared to real squalls I have been in.”

The simulation also scored highly on questions designed to assess its perceived educational value. Overall, participants reported that the simulation did help them feel more prepared if they are caught in a snow squall in the future, and participants reported that it would change their response to a snow squall in the future. One participant from survey group 1 noted “Now, if a snow squall happens I am more prone to get off the next exit due to the possibility of a pile-up and the dangers if driving with children, etc.” Members of survey group 2 similarly remarked “It taught me a lot. I learned how to be prepared for this,” and “I learned what I should have done and how I can avoid the situation getting worse.”

Interestingly, there was a statistically significant difference in the responses between the survey groups to both questions used to assess the perceived educational impact of the simulation. In both the first and third Likert-scale questions, survey group 2 rated the educational aspects of the simulation higher than survey group 1 (Table 1). This aligns with our expectation that participants from diverse backgrounds with less experience driving in local meteorological hazards would find comparatively greater perceived educational value in the simulation, or as one participant put it, “the opportunity to practice driving in a snow squall.” Indeed, survey group 1 participants reported living in an area regularly receiving snow for an average of 32 years, while members of group 2 reported an average of only 8 years, a significant difference. Moreover, 93% of survey group 1 members reported possessing a driver’s license, while only 50% of survey group 2 reported the same, another significant difference. As a result, group 2 survey participants were likely predisposed to find the simulation more educational and impactful, which aligns with the findings of Rickard et al. [42]. In that study, the authors determined that visualizations of another severe weather hazard, tropical cyclone storm surge, were more effective at conveying risk to individuals with less prior experience.

Importantly, since surveying activities spanned a relatively small geographic area encompassing only New York and Pennsylvania, a nuanced approach is necessary when interpreting the survey results. The geographic scale over which the study was conducted means that the results may not be representative of the entire country, or even other snow squall-prone regions. Pennsylvania and New York are two major snow squall hotspots. Therefore, knowledge of snow squalls and familiarity with warning systems prior to using the simulation is likely as high as or higher than other parts of the U.S. As such, including a more geographically diverse sample size would likely produce lower levels of prior knowledge of snow squall hazards. The prior exposure of many participants to snow squalls strengthens the results-based conclusion that the simulation adequately represents the real risks associated with this hazard. However, it is possible that the self-reported impact of the simulation on other populations may differ from the findings of this study. Future work, therefore, could encompass a broader geographic area, or focus on another more constrained region. The latter approach would permit a more direct spatial comparison in simulated snow squall response.

Of note, during the driving scenario the only simulated element that differed linguistically between the two simulation language settings was the WEA message displayed on the in-vehicle mobile phone. The core visual and driving components of the simulation were the same across languages. This suggests that much of the impact of the experience arose from experiential, rather than text-based, communication. Although our study serves as a comparison between groups of participants that spoke English or Spanish, these findings allude to the idea that experience-based VR scenarios may be effective for audiences who speak other languages as well. New York and Pennsylvania have large and diverse non-English speaking communities that are equally affected by snow squalls. While the simulation developed in this work only includes a Spanish translation, the experiential learning aspects would likely hold significant value amongst speakers of other languages even without translation. More broadly, the concepts outlined in this work could be expanded to other meteorological phenomena which are difficult to translate into other languages, providing a more immediate sense of understanding of associated hazards.

4.2 Comparison to Traditional Snow Squall Messaging

Previous snow squall education efforts have relied primarily on passive, text-based information, websites, and outreach campaigns. This initiative, which has only begun over the last several years, is outlined in Section 1.3. Visual depiction has remained a barrier to these efforts, and asking the public to visualize whiteout conditions, poor visibility, and flash freezing of roadways is no substitute for the lived experience.

In contrast to these traditional methods of communication, the VR snow squall simulation attempts to provide education through the creation of a visceral experience. Rather than simply reading about visibility loss, participants visually experience how quickly a snow squall can reduce visibility on the road and the resulting difficulties of driving in such conditions. Our results suggest that the experiential component is valuable for risk communication. Participants registered the threat posed by snow squalls and reported intent for future behavior change.

However, traditional methods of outreach retain the advantages of scalability and accessibility over VR-based solutions. A text message, social media post, or webpage can potentially be shared and viewed by millions of people with no associated hardware cost, whereas VR applications require a specialized headset to experience, limiting their impact to small outreach efforts and individuals who already own the requisite hardware. Additionally, individuals who may be physically unable to don a VR headset would also not be able to experience the simulation. For these reasons, a hybrid approach is worth considering when possible, such as a web browser build, which trades immersion for accessibility. Though not covered in this paper, our team also developed a WebGL browser version of the snow squall simulation after the completion of this study to further expand outreach efforts.

4.3 Improvements Based on Feedback

Some of the qualitative feedback given as part of the free-response questions provided actionable insight into improving the simulation. After the conclusion of the user study, we utilized these critical remarks, as well as our observations from surveying, to adjust the simulation. Specific comments left by participants and the corresponding changes made to improve the simulation can be found in Table 5. All surveys included in the data were conducted using the version of the simulation described in Section 2. Participants did not have access to the improvements made to the simulation described in this section.

Several comments indicated that the participants did not understand what actions they should take to avoid an accident if a snow squall is encountered on the road. One example of such a comment, found in Table 5, was “The VR didn’t really show me what to do if I was caught in a snow squall, it was more of just what would happen.” Though the simulation did include a slide displayed at the end describing actions that should be taken in such a scenario, we noticed while surveying that many of the participants did not notice this slide pinned in front of them to the virtual wall. It was common for users to skip this part of the experience by pressing buttons on the controller quickly, or by immediately taking off the headset after crashing. To reduce the incidence of such actions, we made two changes to the end of the experience. First, we added a delay of several seconds before a user is allowed to return to the start of the simulation. This ensures that the informational slide is not accidentally skipped. Second, we added a voice-over in both English and Spanish that serves as an outro. These audio clips play when the user meets an exit condition (explaining what happened) and accompany the final slide, providing actionable feedback. The explicit guidance relayed by the audio recordings should help some users better understand what to do if caught in a snow squall.

Many participants struggled with the driving controls, examples of which can be found in Table 5. This was the most consistent pain point observed while conducting studies in all locations and among all groups. Though our research team did sit with participants and help them with the controls, much of the negative feedback left in the post-survey centered on these challenges. We made several changes to the simulation based on our observations to reduce friction between the user and the simulation. Participants commonly had trouble understanding the need to hold down the side grip buttons on the Quest 3 Touch controllers to grab onto the wheel. Frequently, participants failed to realize that they were not holding the wheel or accidentally let go of the grip buttons when accelerating. Both caused them to be unable to control the vehicle's steering. To improve this aspect, we focused on making it more obvious when the wheel was not properly grabbed through the usage of visual cues. The final version of the simulation features updated materials for the hand models, modifying them from a neutral color in the survey version to a dark red in the finished application. The hand material in the final version also changes to a bright blue when that hand is gripped onto the steering wheel, providing clear visual feedback that the user can now steer the vehicle. Additionally, we display a message floating on the windshield if the user has no hands on the steering wheel, letting them know that they are not operating the simulated vehicle properly. We also implemented a button that quickly resets the vehicle's position, which the user can use should they immediately get stuck when learning the driving controls. This is designed to eliminate the scenario where a user loses control of the vehicle due to unfamiliarity with the steering mechanics is unable to realign their vehicle back onto the road without restarting the simulation.

Table 5 Selected comments left by participants in the free-response questions and changes made accordingly.

Participant feedback	**Description of change**
● "It helps me be aware of it more but doesn't change how I'll deal with it" ● "The VR didn't really show me what to do if I was caught in a snow squall, it was more of just what would happen"	● Added voice over to the end of the simulation explaining what to do.
● "I feel like I still had pretty decent visibility in the snow squall" ● "Squall could have been even less visibility"	● Reduced the visibility by moving the simulated fog box closer to the vehicle.
● "My experiences of driving in squalls was much more valuable than playing a "game". The VR was not as comfortable as driving." ● "I've been in worse snow squalls, but it was a reasonable showing on the VR. Car controls were challenging."	● Changed the controls to make the car easier to handle. ● Moved reverse to a different button from brake. ● Colored hands to indicate the steering wheel was grabbed.

● "I didn't find it overly helpful since I was so focused on the VR driving mechanics."	● Made it easier to grab on to the steering wheel. ● Added a message instructing the user to grab the steering wheel. ● Added a button to reset the vehicle's position.

4.4 Comparison to Previous Driving Studies

The value of driving simulators in driver training and crash rate reduction has been widely demonstrated [43–45]. VR simulations also have demonstrated effectiveness in complex skills training [10, 46, 47]. Therefore, the use of VR technology for training, risk avoidance, and general education as it relates to driving warrants exploration. Though much research has been done in the space of driving simulators, considerably less has focused on VR-specific simulations, likely due to consumer VR's relative novelty [48, 49].

The simulation outlined in Backlund et al. [50] focused on creating a realistic car for users to sit in but did not mention the addition of agents that replicate real world behavior. Similar to ours, other simulations have sought to include non-player agents that mimic traffic and interact with the user to enhance learning [51–53]. None of the aforementioned efforts utilized VR to create an easily accessible and more immersive experience. While Lang et al. [54] integrated VR technology into a driving simulation, their work focused solely on human-created hazards and general driving habits in everyday circumstances. In contrast, our work focused on simulating weather hazards in the context of passenger vehicle operation. While our simulation did include non-player agents representative of other drivers, the primary purpose was to create an environment akin to what might be encountered during a snow squall. Our simulation also focused explicitly on creating an immersive VR experience that immediately demonstrated hazardous driving conditions, as opposed to an application designed to collect quantitative performance metrics. Additionally, the VR snow squall driving simulation does not require the use of hardware beyond the Quest headset, such as a physical steering wheel or pedals as used in [54], to experience. This makes the snow squall driving simulation more accessible and easier for the public to use, increasing its viability as an outreach tool for snow squall awareness. It is also considerably cheaper to experience, as additional accessories do not need to be purchased to use the simulation.

Finally, our study differs in its population focus. There are a multitude of studies that consider the effects of driving simulators across different age groups, with particular attention paid to effects on both younger novice drivers and older adults [47, 55, 56]. However, the effect of simulations on different language groups has not been widely studied. Our study makes this a primary consideration, though there is potential to expand work in this area given that our analysis is limited to driving in snow squalls.

5 Limitations and Future Work

While the simulation developed for this study serves as an attempt to improve understanding of snow squalls and the risks they present to drivers, our evaluation was limited by the use of self-reported survey data. Snow squalls are not terribly common in any one location [57] which compounds their danger. At the same time, however, it is challenging to gather objective driving performance data and measure if the simulated experience has a substantial impact on driver decision-making in real snow squall conditions. In the future, a longer-term longitudinal study could be performed by testing the simulation on the same participants over the course of several months to collect more robust data that could perhaps better serve as an indicator of behavioral changes, though this would still would most likely be limited to a simulated experience.

VR-based simulations are also limited in their ability to influence regions outside of the U.S. and Canada due to lower market adoption [58]. As of now, the Meta Quest line remain the most popular VR headsets, yet are only available in 23 countries [59]. These factors limit the application of the snow squall simulation and other similar VR-based interventions outside of the U.S. Another confounding factor is the preference community leaders have for VR experiences that are customized to and representative of their local area [17], which makes adaption of simulations such as this to other countries a costly endeavor.

The current snow squall simulation depicts the perspective of a passenger vehicle, but potential utility exists in modifying the simulated vehicle to represent a commercial truck. Many snow squall pileups, and disproportionately the ones that cause fatalities, involve commercial vehicles on major interstates [60, 61]. In addition, many truck drivers passing through the northeast may not be familiar with snow squalls and their dangers because Snow Squall Warnings are relatively new, and they hail from other portions of the country where squalls are not as frequent. It could be helpful to partner with trucking associations and/or trucking companies to develop a snow squall driving simulation for truck drivers. Subsequent incorporation of the simulation into the training curriculum would be a worthwhile outcome to pursue as well.

In addition to expanding the snow squall driving simulation, it is also easy to imagine virtual reality simulations for other weather hazards. Driving hazards like dust storms, fog, or super fog (fog + wildfire smoke, which is common in Florida) could be easily simulated. A simulation for storm chasing could be an effective way to visualize supercell and tornado structure, educating the public about how to safely chase a thunderstorm and what visual cues to look for when identifying potentially hazardous weather conditions. Simulations already exist for phenomena such as hurricanes [17, 62] and rip currents [63, 64], but they could be enhanced with the help of more recent technological advances and new rendering techniques such as 3D Gaussian Splatting [65].

While outside of the scope of the current work, snow squalls will continue to present problems for advanced driver-assistance systems (ADAS) and autonomous vehicles (AVs). These increasingly common autonomous systems rely on convolutional neural networks (CNNs) for traffic sign detection (TSD) and roadway detection—a necessity for their functionality [66]. Much like human vision, camera-based vision sensors and detection algorithms are also challenged by adverse weather conditions [67, 68], with snowfall, fog, and changes in lighting conditions having been shown to affect sensor performance and TSD [67, 69]. TSD models are often trained and evaluated on datasets with ideal weather conditions, leaving the impact of inclement weather such as snowfall an area of active research [70]. The occluding snow particles, low visibility, and lighting conditions associated with snow squalls may present a unique challenge for AVs and ADAS that should be accounted for when evaluating these pieces of software in the future. Additionally, ADAS were not incorporated into the present simulation, but may be worth including in the future given their widespread adoption.

6 Conclusion

This paper presents a simulated snow squall driving scenario in immersive VR. The simulation was developed using information from domain experts to create a realistic approximation of a snow squall and the associated roadway dangers. Our application was open-ended, allowing users a large degree of freedom in the actions they can take in the scenario. We surveyed 216 individuals using the simulation across English and Spanish-language cohorts. Our results indicate that the simulated driving experience left participants feeling more aware of the risks caused by snow squalls and better prepared to act should they encounter a snow squall in real life. We also examined the differences in response between both language groups and found that Spanish-speaking cohort had significantly higher positive impressions of the simulation.

When designing the simulation, we took great pains to capture the look and feel of a snow squall and ensure scientific accuracy. The very spirited responses collected during the user study indicate that the simulated scenario was impactful, and users are likely to remember the experience going forward. Ensuring the delivery of correct information is therefore of paramount importance. This insight must be kept in mind when designing future simulations for the purpose of public safety training.

The efforts taken to design and implement this simulation lay out a foundational framework for future work. For weather hazards in general, it is desirable for individuals to be able to visualize and "experience" a hazard without actually encountering the associated risks in real life. NWS offices and on-air meteorologists go to great lengths to describe the expected impacts of a weather event, including dangerous travel conditions in winter weather, flying debris in tornadoes, or structural damage from hailstorms. With significant advancements in VR technology, visualizing weather hazards is easier—and perhaps more imperative—than ever. Whereas local newscasts and the weather forecasts therein used to be a non-negotiable part of a daily routine, the advent of streaming platforms, on-demand entertainment, and social media necessitate visually appealing, entertaining, and scientifically accurate depictions of hazardous weather. Without a common voice to convey imminent hazards, it is of critical importance that greater efforts be taken towards wide availability of safety information in both traditional and up-and-coming methods of media. VR provides a powerful visualization tool to facilitate this goal.

In classical risk communication, gain versus loss framing is often used to prescribe what types of messaging should be used to influence desired behavior. Prescription commercials depict energetic and joyful actors having a great time with friends and family to encourage viewers to participate in active and healthy lifestyles. On the flip side, a powerful tool to discourage viewers from participating in an activity that could hurt them is on display with AdCouncil messages encouraging people to stop smoking or using tobacco. The raspy voice and scarred throat provide a striking reality of the consequences of harmful actions. As it relates to weather hazards, messaging centers on either (1) engaging in activities that will help you (i.e. when thunder roars, go indoors) or (2) stopping activities that could harm you (i.e. when there is flooding, turn around, don't drown). In the latter case, depicting the consequences of improper actions is powerful. Meteorologists regularly share videos of people driving into flooded roadways, only to see their car swept down a raging river. The goal of sharing these videos is to discourage irresponsible behavior and demonstrate dangerous consequences.

Deploying VR for weather scenarios opens a promising opportunity to illustrate the dangers associated with weather hazards while posing no actual threat to life or property. The snow squall VR simulation allows participants to experience the rapid reduction in visibility, encounter real-life scenarios like a disabled vehicle and a driver zipping by in the left lane, and attempt to control a vehicle while approaching a pileup on an icy roadway. Watching a video of a pileup certainly elicits an emotional response, but the mental process of navigating the situation in real-time almost certainly will do a better job of fostering behavior change in the future.

The intersection between VR and weather holds great potential for future work. Meteorologists remain deeply committed to educating the public about weather hazards with the goal of protecting lives and property. Meanwhile, VR technology continues to advance such that realistic depictions of weather hazards are within reach at more affordable rates in accessible formats. This snow squall driving simulation represents a successful pilot of this intersection and is worthy of additional exploration in the future.

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